

Notation & Definitions

Hat notation: $\hat{v}$ is the unit vector along $\vec{v}$:

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|}, \quad |\hat{v}| = 1$$

Vector components (Cartesian):

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z} = A_i \hat{e}_i$$

A_i is the i -th component ($A_1 = A_x$, etc.).

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{A_i A_i}$$

Index Notation (Einstein Summation)

Repeated index $\Rightarrow$ sum over 1, 2, 3:

$$a_i b_i \equiv \sum_{i=1}^3 a_i b_i$$

Free indices must match on both sides of any equation.

Kronecker Delta

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad \delta_{ij} = \delta_{ji}$$

Properties:

$$A_j \delta_{ij} = A_i \quad \delta_{ik} \delta_{jk} = \delta_{ij}$$

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij}$$

Levi-Civita Symbol

$$\varepsilon_{ijk} = \begin{cases} +1 & (i, j, k) \text{ cyclic} \\ -1 & \text{anti-cyclic} \\ 0 & \text{repeated index} \end{cases}$$

Cyclic: (1, 2, 3), (2, 3, 1), (3, 1, 2) Anti: (2, 1, 3), (1, 3, 2), (3, 2, 1)
Swapping any two indices flips sign:

$$\varepsilon_{ijk} = -\varepsilon_{jik}$$

Key identity:

$$\sum_k \varepsilon_{ijk} \varepsilon_{mnk} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}$$

Dot Product

$$\vec{A} \cdot \vec{B} = AB \cos \theta = A_i B_i$$

Cross Product

$$|\vec{a} \times \vec{b}| = ab \sin \theta \quad \vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

Determinant form:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

Index form (component i of $\vec{C} = \vec{A} \times \vec{B}$):

$$C_i = \varepsilon_{ijk} A_j B_k$$

Cartesian basis:

$$\hat{x} \times \hat{y} = \hat{z}, \quad \hat{y} \times \hat{z} = \hat{x}, \quad \hat{z} \times \hat{x} = \hat{y}$$

$$\hat{y} \times \hat{x} = -\hat{z}, \quad \hat{z} \times \hat{y} = -\hat{x}, \quad \hat{x} \times \hat{z} = -\hat{y}$$

$$\hat{x} \times \hat{x} = \hat{y} \times \hat{y} = \hat{z} \times \hat{z} = \vec{0}$$

Key Identity

BAC-CAB rule ($(\vec{A} \cdot \vec{C})$ and $(\vec{A} \cdot \vec{B})$ are scalars):

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$$

Polar Coordinates (r, θ)

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right)$$

Unit vectors (position-dependent!):

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

$$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta} \quad \hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$

Spherical Coordinates (r, θ, φ)

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \varphi = \arctan\left(\frac{y}{x}\right), \quad \theta = \arccos\left(\frac{z}{r}\right)$$

Unit vectors (position-dependent!):

$$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$$

$$\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$$

Inverse:

$$\hat{x} = \sin \theta \cos \varphi \hat{r} + \cos \theta \cos \varphi \hat{\theta} - \sin \varphi \hat{\varphi}$$

$$\hat{y} = \sin \theta \sin \varphi \hat{r} + \cos \theta \sin \varphi \hat{\theta} + \cos \varphi \hat{\varphi}$$

$$\hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$